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15 NOVEMBER 1977
SUPERSEDING
MIL-P-60363 (MU)
24 February 1966

MILITARY SPECIFICATION

PROPELLANT, M9
(FOR USE IN BASE CHARGE ASSEMBLIES)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M9 Propellant for use in M494A4, APERS-T, Cartridge, 105MM.

1.2 Classification. The propellant shall be M9 flake.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.3)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9230008 - Ejection Charge, Loading, Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision)).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M9 Flake, conforming to the requirements of MIL-STD-652 and Drawing 9230008.

3.2 Graphite (Glaze). The graphite content shall not exceed 0.2 percent when determined as specified in 4.4.3.

3.3 Closed Bomb (Relative Quickness and Relative Force). The M9 propellant shall produce a relative quickness and a relative force of 97 percent (%) to 102% when tested as specified in 4.4.2.

3.4 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality conformance inspection

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and testing.

4.3.2.1 Sampling (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652.

4.4 Test methods and procedures

4.4.1 Chemical Testing. The M9 propellant shall be tested in accordance with Standard MIL-STD-652.

4.4.2 Closed Bomb. The relative quickness and relative force shall be determined in accordance with method 801.1 of Standard MIL-STD-286 with the following exceptions: Propellant lot RAD-64666, M9 for 81MM, M374A2, with a thickness of 0.022 inch shall be used as the reference standard and the test temperature shall be 90 ± 2°F. In addition, the following test parameters are specified:

Volume of Bomb - 200 cc
Charge Weight - 15.784 grams
Ignition Material - 110 grams Ethylcellulose (EC)
Blank Fire - 0.1 grams Nitrocellulose
Loading Density (including ignition material) - 0.084 gm/cc
Sequence of Firing - 3 shots of standard
3 shots of test propellant fired alternately.

4.4.3 Graphite. The graphite content shall be determined in accordance with Method 308.1 of Standard MIL-STD-286.

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5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078(PA) with the following exceptions.

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078(PA) shall apply except for First Article Sample.

6.1.2 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of the Propellant Description sheet shall be forwarded to: Commander, US Army Armament R&D Command, ATTN: DRDAR-QAR-R, and one copy to ATTN: DRDAR-LCU-EP, Dover, New Jersey 07801.

6.2 Surveillance sample. Samples shall be forwarded to US Army R&D Command, Dover, New Jersey, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1376-A088

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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